

Screening for type 2 diabetes ? WHO and IDF report

1 Introduction

Over the past decade the prevalence of type 2 diabetes has increased rapidly. WHO predicts at least 350 million people with type 2 diabetes worldwide by 2030. Around half of people with diabetes are undiagnosed. This report focuses on screening for type 2 diabetes in non-pregnant adults.

2 Background & Rationale for Screening

Diagnostic criteria: Fasting plasma glucose (FPG) ≥ 7.0 mmol/L (126 mg/dL) or 2-h post-load glucose ≥ 11.1 mmol/L (200 mg/dL).

Impaired glucose tolerance (IGT) and impaired fasting glycaemia (IFG) are intermediate risk categories for diabetes and CVD.

Screening rationale:

? Long asymptomatic period during which condition can be detected.

? Substantial proportion of undiagnosed cases.

? Evidence supporting efficacy of blood glucose, blood pressure, and lipid control in reducing complications.

3 Aims of Screening Policy

To review scientific evidence for screening, make health policy recommendations, and guide early detection of type 2 diabetes and pre-diabetes.

4 Terminology & Types of Screening

Definition: "Screening is the process of identifying those individuals who are at sufficiently high risk of a specific disorder to warrant further investigation or direct action."

Types of screening:

1. Selective or targeted screening: Performed in subgroups at high risk (age, obesity, ethnicity, family history).
2. Opportunistic screening: Carried out when individuals attend health care for other reasons.
3. Haphazard screening: Lacks coherent policy (discouraged).

5 Evaluating Screening Tests & Performance Indicators

Key statistical indicators:

- ? Sensitivity: Proportion of true positives detected.
- ? Specificity: Proportion of true negatives correctly identified.
- ? Positive Predictive Value (PPV): Probability that a positive test indicates true diabetes.
- ? ROC Curve: Trade-off between sensitivity and specificity across cut-points.

Screening methods evaluated:

- Risk questionnaires (e.g. Finnish Diabetes Risk Score FINDRISC)
- Fasting plasma glucose (FPG cut-point 5.6 - 6.1 mmol/L)
- Oral glucose tolerance test (OGTT)
- HbA1c and capillary glucose meters.

6 Evidence Base, Prevention & Economic Considerations

Lifestyle Interventions (DPP, Finnish DPS, Da Qing studies):

- ? Intensive lifestyle modifications (diet, exercise) reduce diabetes incidence by 58% in people with IGT.
- ? Metformin reduces incidence by 31%.

Case Studies in National Screening:

- Brazil: Nationwide community screening campaign tested 21.8 million adults aged 40+, identifying 0.25 million severe cases and 0.61 million for confirmatory diagnostic testing.
- Mexico: Two-phase screening using "You have diabetes but you don't know it" questionnaire followed by capillary glucose testing.

Conclusions & Recommendations:

1. Opportunistic targeted screening of high-risk adults is justified if health systems have capacity for diagnostic confirmation and clinical management.
2. Screening programmes should combine risk factor assessment with biochemical testing (FPG / OGTT).
3. Early detection must be linked to structured lifestyle and pharmacological prevention.